Existing water-tracking applications have several limitations:

1. Many applications use fixed daily water targets.
2. Environmental factors such as temperature and humidity are often ignored.
3. Users receive reminders without understanding their actual hydration progress.
4. Historical hydration data is not always converted into meaningful insights.
5. Water wastage is generally not tracked alongside water consumption.
6. Campus water infrastructure problems do not have a dedicated reporting workflow.
7. AI features in many applications are limited to generic chatbot functionality.

Therefore, there is a need for a system that combines personal hydration intelligence with water sustainability and real-world infrastructure reporting.

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3. Proposed Solution

AquaSense provides a centralized mobile platform where users can:

* Create a personalized hydration profile.
* Receive an estimated daily hydration target.
* Log water consumption.
* Receive weather-aware recommendations.
* Monitor hydration progress.
* View an app-generated hydration risk indicator.
* Analyze weekly and monthly hydration patterns.
* Record avoidable water wastage.
* Monitor water-use efficiency.

* Report water-related campus problems.

* Track submitted reports.

* Receive AI-generated insights based on historical data.

The system is designed around a modular architecture so that additional technologies such as IoT-based water-flow sensors can be integrated in future versions.

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4. Major Modules

4.1 User Authentication and Profile

The application provides user registration and login functionality. Users can create profiles containing information required for personalization, such as weight, activity level and daily routine.

Authentication and authorization are handled securely through the backend.

4.2 Personalized Hydration Engine

Instead of using one fixed target for every user, AquaSense generates an estimated hydration requirement using configurable factors such as:

* User profile

* Activity level

* Daily routine

* Environmental conditions

The calculation is deterministic and transparent so that the user can understand why the recommendation changed.

4.3 Weather-Aware Recommendation

The application can integrate with a weather service to retrieve environmental information such as temperature and humidity.

The information can be used to adjust the application's hydration recommendation.

For example:

Base estimated requirement: 2.4 L

Environmental adjustment: +0.3 L

Activity adjustment: +0.2 L

Estimated daily target: 2.9 L

These values are application estimates and are not intended to replace professional medical advice.

4.4 Hydration Risk Analytics

The system compares expected hydration progress with actual logged intake.

For example:

Expected intake by 4 PM: 1.8 L

Actual intake: 1.1 L

Current gap: 0.7 L

The application can convert these factors into an understandable Hydration Risk Score and display the result using a low, moderate or high status.

The score is intended as a wellness indicator rather than a medical diagnosis.

4.5 Water Consumption and Analytics

Users can quickly record water intake using predefined quantities or a custom amount.

Historical data is stored and visualized through:

* Daily statistics
* 7-day trends
* 30-day trends
* Target achievement
* Average intake
* Consistency
* Risk history

This allows users to understand long-term behavior rather than only viewing today's consumption.

4.6 Water Waste Intelligence

AquaSense extends beyond hydration by tracking avoidable water wastage.

Users can record:

* Container quantity
* Consumed quantity
* Remaining quantity
* Reason for wastage

The application can calculate:

Water Waste = Total Water Available − Water Consumed

The resulting data is presented through daily and weekly sustainability metrics.

4.7 Campus Water Issue Reporting

This module addresses a practical institutional problem.

Students can report:

* Water leakage
* Broken taps
* Damaged dispensers
* Water availability problems
* Water-station issues
* Other water-related infrastructure problems

Reports may contain:

* Category
* Location

* Description
* Photograph
* Priority
* Date and time

Administrators can manage reports through statuses such as:

Submitted → Assigned → In Progress → Resolved

This creates a basic digital workflow between students and campus maintenance.

4.8 AI-Powered Insights

The AI module is designed to analyze actual application data rather than functioning as a generic chatbot.

Examples of possible insights include:

* Consistently missed hydration periods
* Changes in weekly hydration performance
* Water-waste trends
* Target achievement patterns

AI-generated information is presented as an insight layer, while the core calculations remain deterministic and testable.

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5. Technology and Architecture

The proposed application uses Flutter and Dart for cross-platform mobile development.

The backend can be developed using Node.js and Express.js, while MongoDB can be used for persistent data storage.